Gear Exciter

User manual and parameter reference · 0.1.1

An exciter combining level-dependent low/high shaping, harmonics and adjustable band timing.

Quick start

1. Insert Gear Exciter on an audio track and start with moderate settings.
2. Compare Process/Contour with Harmonics to hear the different stages.
3. Adjust Focus and Colour, compare at equal level, and use HELP for the current control.

Editor window

1200 × 548 fixed. Dimensions are logical editor pixels, before host window chrome. HELP adds 92 pixels of height (Gear adds a one-pixel divider too). Allow enough usable screen space at your display scale. Press HELP again to close the help area.

Operation

Output follows the blend, so it changes the complete mixed signal. Process and Contour adjust existing bands; the harmonic stage creates and adds new harmonics.

Arm Auto Gain before a direct sound-knob edit to request a short learning period. Switches, automation, presets and session recall do not request learning.

Auto Gain is an armed workflow, not continuous normalization. Only direct user changes to sound knobs start learning while it is enabled; it holds for eight seconds after the last such edit and then idles. Output is excluded so you can trim it while idle. Turning Auto Gain on only arms it. Switching it off stops learning. Presets do not store Auto Gain; session recall retains the switch and Output but starts idle. Offline rendering uses the saved output.

Controls

- **Process**: lifts the highs (above 1.2 kHz) by an amount that follows their level, up to +12 dB on quiet detail and +2 dB on loud passages, so transients are not harshened. 0 to 10.
- **Contour**: the same for the lows (under 150 Hz), up to +9 dB when quiet and +1.5 dB when loud, so the bottom thickens without booming. 0 to 10.
- **Align**: holds the lows back by up to 0.6 ms and the mids by up to 0.15 ms relative to the highs. At 0 the bands sum flat and nothing is changed.
- **Harmonics**: how much of the harmonics made from the highs is added. 0 adds nothing.

- **Focus**: where the harmonic stage starts, 1.5 to 8 kHz: only what is above it is saturated, and only harmonics above it are kept.
- **Colour**: Warm leans to even harmonics (an asymmetric curve), Bright to odd ones.
- **Double Trouble** (top bar): doubles Process and Contour's boosts in dB and doubles the added harmonic signal (+6 dB), at the same knob settings. The maximum quiet-band boosts become +24 dB on the highs and +18 dB on the lows. The switch ramps over 20 ms; Align keeps its timing. Off is the original strength. Saved with presets and sessions; older ones load with it off. It does not start Auto Gain; only direct sound-knob edits request matching.
- **Auto Gain** (top bar): arms loudness matching. Only a direct sound-knob edit starts it turning Output (-12 to +12 dB) to match the dry input, then it idles with the knob where it was. Other switches, automation and preset/session loads do not start a new match.
- **HELP** (top bar): a bar under it says what the control under the mouse is and what it does.
- **Mix**, **Output** (-12 to +12 dB), **Link** (the channels follow one level so the image does not lean), **Quality** (the harmonic stage's oversampling, 8x or 16x), **Bypass**.

The plugin uses one reported latency for both quality settings (about 1 ms at 48 kHz), so quality changes do not move the track. The dry signal, Mix and Power/host bypass use the same delay. Power fades over 5 ms, and the added-harmonics path fades around a quality change while its filters settle.

Controls and operation

PROCESS

Lifts the highs (above 1.2 kHz) by an amount that follows their level, 0 to 10: up to +12 dB on quiet detail, down to +2 dB on loud passages. Double Trouble doubles these boosts to +24 and +4 dB. The air and the edge of a sound come forward; the lift moves with the music.

CONTOUR

Fills out the lows (under 150 Hz) by an amount that follows their level, 0 to 10: up to +9 dB when they are quiet, +1.5 dB when they are loud. Double Trouble doubles these boosts to +18 and +3 dB.

ALIGN

Holds the lows back by up to 0.6 ms and the mids by up to 0.15 ms against the highs, 0 to 100%, to change their phase relationship. At 0 the three bands add up flat; increasing Align changes the timing rather than adding an EQ boost.

HARMONICS

How much of the harmonics made from the highs is added, 0 to 100%. The part of the signal above Focus is saturated and what the saturation makes (new harmonics above the notes, not more of the notes) is added back. Quiet material gets the same proportion as loud. Double Trouble doubles the added signal (+6 dB); 0 adds nothing in either mode.

FOCUS

Where the harmonic stage starts, 1.5 to 8 kHz: only what is above it is saturated, and only harmonics above it are added. Low is a fuller presence; high is only air.

COLOUR

The kind of harmonics: Warm leans to even ones (2nd, 4th: a lean, tube-like curve), Bright to odd ones (3rd, 5th: a symmetric, harder edge), Centre between them.

MIX

Both stages together, the time alignment and the harmonics, against the dry signal, 0 to 100%, before Output. Under 100% the held-back lows add up with the dry signal's and comb a little, by design; the dry signal is delayed by the plugin's latency so the two line up.

OUTPUT

The level after both stages and the mix, -12 to +12 dB. Compare with Power off to judge the sound and not the loudness, or let Auto Gain set it.

DOUBLE TROUBLE

Twice the excitation: doubles Process and Contour's boosts in dB (up to +24 dB on quiet highs, +18 dB on quiet lows) and the added harmonic signal (+6 dB). Ramps over 20 ms; Align keeps its timing. Saved with presets/sessions; off is the original strength. Switches, including Double Trouble, do not start Auto Gain matching.

AUTO GAIN

Arms loudness matching. Only a direct sound-knob edit starts it turning Output (-12 to +12 dB) to match the dry input (K-weighted, over 1.2 s), then it goes idle. Output edits, switches, automation and preset/session loads do not start a match. Working shows as a pulsing dot and a line running down the time left; idle as a steady dot. Switch it off and the knob keeps its value.

LINK

Stereo link. On, both channels follow the same levels, so Process and Contour treat them alike and the image does not lean. Off, each channel follows its own.

QUALITY

The harmonic stage's oversampling: 8x (up) or 16x (down), both with anti-aliasing of the curve. 16x keeps the harmonics from folding back as aliasing a little further at more CPU. Both settings share the same reported latency and the added harmonics fade around switching. Click to switch.

VU METER

The output level. 0 VU is -18 dBFS RMS; the needle follows the signal's average as a VU meter does, full scale at +3 VU. It rests when Power is off.

POWER

Switches the exciter on or off with a short fade. Off plays the input delayed by the same reported latency.

POWER LAMP

Lit while the exciter is on.

PRESETS

Browse, load and save settings. An asterisk means the sound has changed since the preset was loaded.

HELP

Shows or hides this bar. It says what the control under the mouse is and what it does.

BYPASS

The same as the POWER switch: a short fade to the latency-aligned input.

Every parameter

Values and defaults are taken from the plugin. A control's effect can depend on the selected mode; see the operation notes above. Choices keep their saved identity between versions. Knobs and switches can be automated unless marked otherwise.

Parameter	What it does	Values and default
Process	Process: 4.0	0.0 to 10.0 Default: 4.0
Contour	Contour: 3.0	0.0 to 10.0 Default: 3.0
Align	Align: 100%	0% to 100% Default: 100%
Harmonics	Harmonics: 30%	0% to 100% Default: 30%
Focus	Focus: 3.00 kHz	1.50 kHz to 8.00 kHz Default: 3.00 kHz

Parameter	What it does	Values and default
Colour	Colour: Centre	Warm 100 to Bright 100 Default: Centre
Output	Output: 0.0 dB	-12.0 dB to +12.0 dB Default: 0.0 dB
Mix	Mix: 100%	0% to 100% Default: 100%
Link	Link: On	Off to On Default: On
Quality	Quality: 8x	8x, 16x Default: 8x Not host automatable
Auto Gain	Arms loudness matching: only direct sound-knob edits start eight seconds of learning, then it idles; switches, automation and preset/session loads do not start it	Off to On Default: Off
Bypass	Plays the input as it is (the POWER switch on the device)	Off to On Default: Off
Double Trouble	Twice the Process and Contour boost in dB and twice the added harmonics; Align, Mix and Output keep their settings	Off to On Default: Off

Presets and files

Presets are browsed with the arrows or the preset-name menu. A star marks an edited patch. Save supports folders, Rename, Delete and opening the user preset folder; dropping a compatible preset file onto the bar loads it. A newer-file indication means this version could not apply every setting.

The host session stores the complete patch. A product preset can deliberately leave listening trim, audition or display controls unchanged; the product-specific rules below describe those exceptions.

Gear Exciter presets use .gearexciter in ~/Library/Audio/Presets/Barakigal Audio/Gear Exciter. Power/Bypass and Auto Gain are excluded from presets; Output belongs to the processing sound and is included.

Installation and help

Download the macOS archive from the product page and quit your audio host. Place the AU .component in ~/Library/Audio/Plug-Ins/Components, VST3 in ~/Library/Audio/Plug-Ins/VST3, or CLAP in ~/Library/Audio/Plug-Ins/CLAP. The product page also offers an installer with format selection. Restart or rescan the host. To uninstall, quit the host and remove that product's bundles from those folders; saved sessions and user presets are separate.

For questions or a problem report, use BRK Audio support on GitHub. Include the plugin version, macOS version, host and format, sample rate and a short description of how to reproduce it.